

Cores of Binary 3-Qualitative Independence Hypergraphs

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Abstract

Let $3\text{-QI}(n, 2)$ be the 3-uniform hypergraph whose vertices are binary bipartitions of an n -set with both classes of size at least 4, and whose edges are triples realizing all eight binary patterns. We prove that $3\text{-QI}(n, 2)$ is a core for every $n \geq 8$. The proof has two parts. First, we prove that every uniform middle layer $3\text{-UQI}(n, 2)$ is a core for even $n \geq 8$. The cases $n \equiv 2 \pmod{4}$ are reduced to the nonexistence of a Steiner system, while the cases $n \equiv 0 \pmod{4}$ are handled by a two-anchor critical-fold obstruction and a Sperner bound on link cliques. Second, we prove a middle-layer rigidity theorem: if the most balanced layer is a core, then the full qualitative-independence hypergraph is a core. This uses an exact crossing-family bound, endomorphism invariance of the middle layer, a finite-order power argument, and a balanced-anchor separation construction.

Keywords. Qualitative independence; hypergraph core; graph homomorphism; covering array; folded Johnson graph; Steiner system; Sperner theorem.

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1 Introduction

A finite hypergraph is a *core* if every endomorphism is an automorphism. Cores are the minimal representatives of homomorphic equivalence classes and retain homomorphism-monotone invariants. Qualitative independence hypergraphs encode covering arrays: a triple of binary columns is adjacent precisely when all eight binary row patterns occur.

Thomas and Akhtar initiated the systematic study of cores of binary 3-qualitative independence hypergraphs, proved the case $n = 8$, and left the determination of the core of $3\text{-QI}(n, 2)$ for $n > 8$ as an open problem [4]. They also related the uniform and almost-uniform middle layers to merged Johnson graphs.

For odd $n \geq 9$, the core property of the almost-uniform middle layer was proved in [2] by determining the endomorphism monoid of the relevant merged Johnson graph. The present paper supplies the even middle layers and then lifts the middle-layer result to the full hypergraph.

Theorem 1.1. *For every integer $n \geq 8$, the hypergraph*

$$3\text{-QI}(n, 2)$$

is a core.

Two intermediate results may be useful beyond the present application.

Theorem 1.2 (Even middle layers). *For every even integer $n \geq 8$,*

$$3\text{-UI}(n, 2)$$

is a core.

Theorem 1.3 (Middle-layer lift). *Let B_n be the most balanced layer of $H_n = 3\text{-UI}(n, 2)$. If B_n is a core, then H_n is a core.*

2 Qualitative independence and cores

Let Ω be an n -element set. A binary string is identified, up to interchanging its two symbols, with an unordered bipartition

$$[A] = \{A, \Omega \setminus A\}.$$

The vertices of $H_n = 3\text{-UI}(n, 2)$ are those bipartitions whose two blocks have size at least 4. Three vertices $[A], [B], [C]$ form an edge when all eight cells

$$A^{\epsilon_1} \cap B^{\epsilon_2} \cap C^{\epsilon_3}, \quad (\epsilon_1, \epsilon_2, \epsilon_3) \in \{0, 1\}^3,$$

are nonempty. This condition is independent of the chosen orientations.

The *middle layer* B_n consists of the bipartitions with block sizes

$$\left\lfloor \frac{n}{2} \right\rfloor, \quad \left\lceil \frac{n}{2} \right\rceil.$$

It is $3\text{-UI}(n, 2)$ for even n and $3\text{-AUUI}(n, 2)$ for odd n .

The 2-section $[H]_2$ of a 3-uniform hypergraph H is the graph on $V(H)$ in which two vertices are adjacent when they lie together in a hyperedge.

Lemma 2.1 (Graph-to-hypergraph transfer). *If $[H]_2$ is a core, then H is a core.*

Proof. Every endomorphism of H is an endomorphism of $[H]_2$. If $[H]_2$ is a core, that map is injective. On a finite vertex set it is bijective. The induced map on 3-subsets is then a permutation, and it maps the finite edge set injectively into itself, hence onto itself. $\square$

3 Crossing cuts and the middle-layer lift

For a t -set T , let $\text{Cross}(t)$ be the graph whose vertices are nontrivial unordered bipartitions of T , two of which are adjacent when all four cells of their common refinement are nonempty.

Proposition 3.1. *For $t \geq 4$,*

$$\omega(\text{Cross}(t)) = q(t) := \binom{t-1}{\lfloor t/2 \rfloor - 1}.$$

Moreover, $q(t+1) > q(t)$ for every $t \geq 4$.

Proof. If $t = 2r$, orient each cut toward a fixed point p . Removing p from the chosen sides gives an antichain on $2r-1$ points, so Sperner's theorem gives the upper bound $\binom{2r-1}{r-1}$. Equality is attained by all r -subsets containing p .

If $t = 2r + 1$, the complements of the oriented traces form an intersecting antichain on $2r$ points. Milner's intersecting-antichain theorem gives $\binom{2r}{r-1}$, attained by all r -subsets containing p [3]. The ratios

$$\frac{q(2r+1)}{q(2r)} = \frac{2r}{r+1} > 1, \quad \frac{q(2r+2)}{q(2r+1)} = \frac{2r+1}{r} > 1$$

give strict monotonicity. $\square$

For $T \in V(H_n)$, let L_T denote the link graph:

$$R \sim_{L_T} S \iff \{T, R, S\} \in E(H_n).$$

Lemma 3.2 (Intrinsic middle layer). *Every endomorphism of H_n maps B_n into B_n .*

Proof. Let the smaller block of T have size t . Restriction to that block maps every link clique injectively into $\text{Cross}(t)$, hence

$$\omega(L_T) \leq q(t).$$

If $T \in B_n$, let $b = \lfloor n/2 \rfloor$. Take a maximum $q(b)$ -crossing family on the b -block. On the other block take a $q(b)$ -element crossing family: when n is odd, this is a subfamily of a maximum $q(b+1)$ -family. Pair the local cuts and join corresponding sides. The resulting global bipartitions form a $q(b)$ -clique in L_T . Their two blocks have size at least 4, including the boundary cases $n = 8, 9$.

An endomorphism maps a link clique of size at least two injectively into a link clique. Thus, for $T \in B_n$,

$$q(b) = \omega(L_T) \leq \omega(L_{f(T)}) \leq q(t(f(T))).$$

Strict monotonicity and $t(f(T)) \leq b$ imply $t(f(T)) = b$. $\square$

Lemma 3.3 (Balanced-anchor separation). *For distinct $X, Y \in V(H_n)$, there exist $R, S \in B_n$ such that*

$$\{X, R, S\} \in E(H_n), \quad \{Y, R, S\} \notin E(H_n).$$

Proof. Write $X = \{X_0, X_1\}$ and $Y = \{Y_0, Y_1\}$. Since $X \neq Y$, some block E of Y meets both blocks of X ; otherwise the two partitions coincide. Put $D = E^c$.

Colour Ω with the four colours 00, 01, 10, 11. Choose one point in each $X_i \cap E$ and colour those two points 11. In each X_i , colour three additional points 00, 01, 10. Every X_i now contains all four colours, while D contains no point of colour 11.

For $n = 2k$, complete the colour multiplicities to

$$(c_{00}, c_{01}, c_{10}, c_{11}) = (2, k-2, k-2, 2);$$

for $n = 2k + 1$, complete them to

$$(3, k-2, k-2, 2).$$

The remaining required multiplicities are nonnegative because $k \geq 4$. Define

$$R = \{x : c(x) \in \{10, 11\}\}, \quad S = \{x : c(x) \in \{01, 11\}\}.$$

Then $R, S \in B_n$, every X -block sees all four joint colours, and $D \cap R \cap S = \emptyset$. This gives the two claimed incidences. $\square$

Proof of Theorem 1.3. Let $f \in \text{End}(H_n)$. By Lemma 3.2, $g = f|_{B_n}$ is an endomorphism of B_n , hence an automorphism. Let d be its finite order and set $h = f^d$. Then h fixes B_n pointwise.

If $h(X) = Y \neq X$, choose $R, S \in B_n$ from Lemma 3.3. Applying h maps the edge $\{X, R, S\}$ to the nonedge $\{Y, R, S\}$, a contradiction. Hence $f^d = h = \text{id}$, so f is bijective and its inverse is the endomorphism f^{d-1} . $\square$

4 Even middle layers: the two modulo four branch

Let $n = 2k$, where $k \geq 5$ is odd, and let

$$H_k = 3\text{-UQI}(2k, 2), \quad G_k = [H_k]_2.$$

The vertices are unordered equipartitions $[A] = \{A, A^c\}$, with $|A| = k$. If $s = |A \cap B|$, the four pair cells have sizes $s, k - s, k - s, s$. A balanced third part splits all four cells if and only if

$$2 \leq s \leq k - 2.$$

For sufficiency, choose its intersections with the four cells to have sizes

$$1, \quad 1, \quad k - s - 1, \quad s - 1.$$

Fix a point in the $2k$ -set. The quotient vertices become the r -subsets of a v -set, where

$$r = k - 1, \quad v = 2r + 1 = 2k - 1.$$

Let Γ join two r -subsets when their intersection has size 0 or $r - 1$. Then

$$G_k = \bar{\Gamma}.$$

Lemma 4.1. Γ is vertex-transitive and $\omega(\Gamma) = r + 2$.

Proof. Fixing an $(r - 1)$ -subset gives a clique of size $r + 2$. For the upper bound, first suppose that a clique contains disjoint A, B . There is one point x outside $A \cup B$. If C is adjacent to both, then

$$|C \cap A| + |C \cap B| + \mathbf{1}_{x \in C} = r,$$

with both intersections in $\{0, r - 1\}$. Hence C is

$$(A \setminus \{a\}) \cup \{x\} \quad \text{or} \quad (B \setminus \{b\}) \cup \{x\}.$$

One set of each type meets in the singleton $\{x\}$, so the two types cannot be mixed when $r \geq 3$. The clique has at most $2 + r = r + 2$ members.

Now suppose that there is no disjoint pair. All pairwise intersections have size $r - 1$. Write

$$A = D \cup \{a\}, \quad B = D \cup \{b\}, \quad |D| = r - 1.$$

Every further member either contains D , or is an r -subset of $D \cup \{a, b\}$. If only the first type occurs, the family is contained in the $r + 2$ sets obtained by adjoining one point to D . If the second type occurs, no set $D \cup \{z\}$ with $z \notin D \cup \{a, b\}$ is compatible with it, and the clique is contained in the $r + 1$ r -subsets of $D \cup \{a, b\}$. This proves the bound. $\square$

Lemma 4.2 (Design implication). *If*

$$\chi(\bar{\Gamma}) = \omega(\bar{\Gamma}),$$

then a Steiner system $S(r - 1, r, 2r + 1)$ exists.

Proof. For a clique $\mathcal{Q}$ and independent set $\mathcal{I}$ of the vertex-transitive graph Γ , orbit averaging gives

$$|\mathcal{Q}||\mathcal{I}| \leq \binom{2r+1}{r}.$$

Assume $\chi(\bar{\Gamma}) = \omega(\bar{\Gamma}) = c$. Since $\omega(\bar{\Gamma}) = \alpha(\Gamma)$, a c -colouring of $\bar{\Gamma}$ and Lemma 4.1 force

$$c = \frac{\binom{2r+1}{r}}{r+2}$$

and every colour class has size $r+2$.

Take a maximum clique $\mathcal{B}$ of $\bar{\Gamma}$. It is an independent set in Γ , so no two of its r -subsets share an $(r-1)$ -subset. Moreover,

$$|\mathcal{B}|r = \frac{\binom{2r+1}{r}}{r+2}r = \binom{2r+1}{r-1}.$$

Thus every $(r-1)$ -subset lies in exactly one member of $\mathcal{B}$. □

Proposition 4.3. *If $k \geq 5$ is odd, then 3-UQI($2k, 2$) is a core.*

Proof. The natural symmetric group is transitive on nonedges of G_k . Cameron and Kazanidis therefore imply that G_k is a core or has a complete core [1, Corollary 2.2]. A finite graph has a complete core exactly when its clique and chromatic numbers agree.

By Lemma 4.2, equality would yield

$$S(k-2, k-1, 2k-1).$$

In such a design the number of blocks through a fixed $(k-3)$ -subset is

$$\lambda_{k-3} = \frac{\binom{k+2}{1}}{\binom{2}{1}} = \frac{k+2}{2},$$

which is not an integer for odd k . Hence the complete-core alternative is impossible, so G_k is a core. Apply Lemma 2.1. □

5 Even middle layers: the zero modulo four branch

Let $n = 4m$ and use the fixed-point model

$$V = \binom{\Omega}{2m-1}, \quad |\Omega| = 4m-1.$$

Three vertices form an edge when the seven finite Boolean cells other than 111 are nonempty. The 2-section joins X, Y when

$$1 \leq |X \cap Y| \leq 2m-3.$$

Thus distinct nonedges have intersection 0 or $2m-2$, and form one orbit under the full S_{4m} -action on equipartitions.

Lemma 5.1 (Critical-fold criterion). *Suppose every link of a vertex-transitive 3-uniform hypergraph has clique number at most M , its distinct 2-section nonedges form one orbit, and a standard nonedge P, Q has $M+1$ auxiliary vertices such that every auxiliary pair is in an edge with P or Q . Then the hypergraph is a core.*

Proof. If an endomorphism identifies P, Q , the auxiliary images are distinct and form an $(M + 1)$ -clique in the link of the common image. Every possible collision is equivalent to this one. $\square$

Proposition 5.2. *For every $m \geq 4$, $3\text{-UQI}(4m, 2)$ is a core.*

Proof. For a link clique $\mathcal{C}$ at T , the traces $\{T \cap R : R \in \mathcal{C}\}$ form an antichain in 2^T . Hence

$$\omega(L_T) \leq M_m := \binom{2m-1}{m-1}.$$

Choose disjoint $K, \{p, q\}, W$ with

$$|K| = 2m - 2, \quad |W| = 2m - 1,$$

and put $P = K \cup \{p\}$, $Q = K \cup \{q\}$. Fix $x \in K$ and distinct $u, v \in W$, and put $O = \{p\} \cup W$.

Partition $A \in \binom{Q}{m-1}$ into the three types

$$x \notin A, \quad x \in A, \quad q \notin A, \quad x, q \in A.$$

Match them bijectively with complement pairs of m -subsets of O of the following respective types:

type	condition	number
0	u, v separated	$\binom{2m-2}{m-1}$
10	p -free representative contains u, v	$\binom{2m-3}{m-2}$
11	p -free representative avoids u, v	$\binom{2m-3}{m-3}$.

The three classes on the A -side have the same respective cardinalities: choose $m - 1$ points while avoiding x , choose x and $m - 2$ further points while avoiding q , or choose x, q and $m - 3$ further points. Thus the typewise bijections exist and together exhaust both sides.

Let B_A be the chosen representative and set $R_A = A \cup B_A$. For distinct A, A' , put

$$s = |A \cap A'|, \quad t = |B_A \cap B_{A'}|.$$

Distinct equal-sized A, A' give $0 \leq s \leq m - 2$. Distinct assigned complement pairs make $B_A, B_{A'}$ neither equal nor complementary, so $1 \leq t \leq m - 1$. In bit order $(Q, R_A, R_{A'})$, the seven required atom sizes are

	000	001	010	011	100	101	110
size	t	$m - t$	$m - t$	t	$s + 1$	$m - 1 - s$	$m - 1 - s$.

They are all positive. Hence distinct central vertices satisfy

$$\{Q, R_A, R_{A'}\} \in E.$$

Set

$$E = \{x, p\} \cup (W \setminus \{u, v\}).$$

The required exceptional atom sizes, in bit order (anchor, E, R_A), are

type	anchor	000	001	010	011	100	101	110
0	Q	1	1	$m - 1$	$m - 1$	$m - 1$	$m - 1$	1
10	P	1	2	$m - 1$	$m - 2$	$m - 1$	$m - 2$	1
11	P	2	1	$m - 3$	m	m	$m - 3$	1.

All are positive for $m \geq 4$. Thus

$$\{E\} \cup \left\{ R_A : A \in \binom{Q}{m-1} \right\}$$

has $M_m + 1$ vertices and satisfies Lemma 5.1. $\square$

5.1 The exceptional value $n=12$

The preceding construction has two zero cells at $m = 3$. We use a separate 11-vertex obstruction. In the model $V = \binom{[11]}{5}$, set

$$P = \{0, 1, 2, 3, 4\}, \quad Q = \{0, 1, 2, 3, 5\}.$$

The eleven auxiliary vertices and their anchor table are given in Appendix A. Every link clique has size at most $\binom{5}{2} = 10$, while identifying P, Q forces an 11-clique in one link. Hence 3-UQI(12, 2) is a core.

Proof of Theorem 1.2. The case $n = 8$ is due to Thomas and Akhtar [4, Theorem 4]. Proposition 4.3 treats $n \equiv 2 \pmod{4}$, the preceding subsection treats $n = 12$, and Proposition 5.2 treats $n \equiv 0 \pmod{4}$, $n \geq 16$. $\square$

Proof of Theorem 1.1. For even n , the middle layer is a core by Theorem 1.2. For odd $n \geq 9$, it is a core by [2]. Apply Theorem 1.3. The case $n = 8$ is already included in the even statement. $\square$

A The $n=12$ critical-fold certificate

The auxiliary vertices are

1	$\{1, 2, 3, 5, 10\}$	2	$\{0, 3, 8, 9, 10\}$
3	$\{0, 2, 7, 9, 10\}$	4	$\{2, 3, 4, 9, 10\}$
5	$\{1, 3, 4, 8, 10\}$	6	$\{1, 2, 4, 7, 10\}$
7	$\{0, 5, 7, 8, 9\}$	8	$\{0, 1, 6, 7, 10\}$
9	$\{3, 4, 5, 7, 9\}$	10	$\{2, 4, 5, 6, 9\}$
11	$\{1, 4, 5, 7, 8\}$		

and the upper-triangular anchor table is

	2	3	4	5	6	7	8	9	10	11
1	P	P	P	P	P	P	P	P	P	P
2		P	Q	Q	Q	Q	P	P	P	P
3			Q	Q	Q	Q	P	P	P	P
4				P	P	P	Q	Q	Q	Q
5					P	P	P	Q	Q	Q
6						P	Q	P	Q	Q
7							Q	Q	P	Q
8								P	P	P
9									P	P
10										P

where an entry P or Q means that the corresponding auxiliary pair, together with that anchor, is a hyperedge. Each assertion is a direct seven-cell check.

B AI assistance and verification status

AI systems were used to generate conjectures, search for finite obstructions, propose symbolic constructions, draft text and code, and perform adversarial audits. The author selected, checked,

and organized the arguments and remains responsible for every claim. The accompanying programs verify finite certificates and symbolic identities but do not constitute peer review or formal proof. This manuscript has not been validated by a qualified human specialist.

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